INCH-POUND
MIL-DTL-1189E(SH)
w/AMENDMENT 1
11 March 2020

SUPERSEDING
MIL-DTL-1189E(SH)
24 January 2018

DETAIL SPECIFICATION

# VALVE, GATE, BRONZE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification describes bronze gate valves for use in shipboard water, oil, gas, or steam services.

1.2 Classification. Gate valves are of the following types and classes (see 6.2):

a. Type I – Rising stem, outside screw, and yoke.

(1) Class 2 – For working pressures not greater than 250 pounds per square inch ($lb/in^2$), 2½ to 12 inches nominal pipe size (nps) (water, oil, gas, or steam).

(2) Class 3 – For working pressures 251 to 400 $lb/in^2$, 2½ to 12 inches nps (water, oil, gas, or steam).

(3) Class 4 – For working pressures not greater than 50 $lb/in^2$, 14 to 46 inches nps (seawater).

b. Type II – Non-rising stem, for working pressures not greater than 50 $lb/in^2$, 14 to 46 inches nps (seawater).

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.3 Part or identifying number (PIN). PINs are for general information only. The acquisition requirements (see 6.2) take precedence. The PINs to be used for valves acquired to this specification are created as follows:

| **M** | **1189** | **-** | **X** | **X** | **XXX** | **/** | **XXX** | **XX** |
|---|---|---|---|---|---|---|---|---|
| Prefix for Military Specification | Specification Number | | Type (see code below) | Class (see code below) | Rating (see code below) | / | Nominal Size (see code below) | Potable water use (see code below) |

PIN Code:

| Type Code | | Class Code | | Rating Code (lb/in$^2$) | | Nominal Pipe Size (nps) | | Potable Water Use | |
|---|---|---|---|---|---|---|---|---|---|
| Type | Code | Class | Code | Rating | Code | Size [1/, 2/] | Code | Category | Code |
| I | R | 2 | 2 | 50 | 050 | 2½ | 025 | Suitable | PW |
| II | N | 3 | 3 | 250 | 250 | 3 | 030 | Not Suitable | (blank) |
| | | 4 | 4 | 400 | 400 | 4 | 040 | | |
| | | | | | | 5 | 050 | | |
| | | | | | | 6 | 060 | | |
| | | | | | | 8 | 080 | | |
| | | | | | | 10 | 100 | | |
| | | | | | | 12 | 120 | | |
| | | | | | | 14 | 140 | | |
| | | | | | | 16 | 160 | | |
| | | | | | | 18 | 180 | | |
| | | | | | | [3/] | : | | |
| | | | | | | 46 | 460 | | |

NOTES:

[1/] Note that sizes 14 to 46 may be type I, class 4 or type II only; see table II.

[2/] Nominal size code consists of nps, with a zero in front for small sizes or a zero at the end for large sizes, ensuring a consistent 3-digit code.

[3/] See table VI for additional sizes for type I, class 4 or type II valves.

Example: M1189-R3400/060PW is for a rising stem gate valve, class 3, 400-lb/in$^2$, 6-inch, suitable for potable water use.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.



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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-PRF-1149 - Gasket Materials, Synthetic Rubber, 50 and 65 Durometer Hardness

MIL-DTL-15024 - Plates and Tags for Identification of Equipment, General Specification for

MIL-P-15024/5 - Plates, Identification

MIL-PRF-20042 - Flanges, Pipe, Bronze (Silver Brazing)

MIL-DTL-22050 - Gasket and Packing Material, Rubber, for Use with Polar Fluids, Steam, and Air at Moderately High Temperatures

MIL-B-24480 - Bronze, Nickel-Aluminum (UNS No. C95800) Castings for Seawater Service

MIL-P-24503 - Packing Material, Graphitic, Corrugated Ribbon or Textured Tape and Preformed Ring

DOD-G-24508 - Grease, High Performance, Multipurpose (Metric)

DOD-V-24657 - Valve Actuator, Direct Coupled, Gear Driven Electrically Powered for Naval Shipboard Propulsion and Auxiliary Systems (Metric)

MIL-C-24723 - Castings, Nickel-Copper Alloy

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)



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NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-1385620 - Handwheels for Valves

803-2177917 - Valves, Bronze, Flanged, Gate 250 PSI WOG

(Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of these documents may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

0900-LP-001-7000 - Fabrication and Inspection of Brazed Piping Systems

S9074-AR-GIB-010/278 - Requirements for Fabrication, Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)/AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ANSI/ASME B1.5 - Acme Screw Threads

ANSI/ASME B18.2.1 - Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)

ANSI/ASME B18.2.2 - Nuts for General Applications: Machine Screw Nuts, Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)

(Copies of these documents are available online at https://webstore.ansi.org.)

ANSI/AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM B61 - Standard Specification for Steam or Valve Bronze Castings

ASTM B98/B98M - Standard Specification for Copper-Silicon Alloy Rod, Bar, and Shapes

ASTM B148 - Standard Specification for Aluminum-Bronze Sand Castings

ASTM B164 - Standard Specification for Nickel-Copper Alloy Rod, Bar, and Wire



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ASTM B584 - Standard Specification for Copper Alloy Sand Castings for General Applications

ASTM D1710 - Standard Specification for Extruded Polytetrafluoroethylene (PTFE) Rod, Heavy Walled Tubing and Basic Shapes

(Copies of these documents are available online at www.astm.org.)

MANUFACTURERS STANDARDIZATION SOCIETY (MSS) OF THE VALVE AND FITTINGS INDUSTRY

MSS SP-91 - Guidelines for Manual Operation of Valves

(Copies of this document are online at www.mss-hq.org.)

NSF INTERNATIONAL/ANSI

NSF/ANSI 61 - Drinking Water Systems Components - Health Effects

(Copies of this document are available online at www.nsf.org.)

SAE INTERNATIONAL

SAE HS 1086 - Metals and Alloys in the Unified Numbering System

SAE AMS4676 - Nickel-Copper Alloy, Corrosion-Resistant, Bars and Forgings, 66.5 Ni - 3.0Al - 0.62Ti - 28Cu Hot-Finished, Precipitation Hardenable

SAE AMS4892 - Alloy Castings, Corrosion and Heat Resistant 66Ni - 29Cu - 4.0Si As Cast

SAE AMS7276 - Rubber: Fluorocarbon (FKM) High-Temperature-Fluid Resistant Low Compression Set for Seals in Fuel Systems and Specific Engine Oil Systems

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Unless otherwise specified (see 6.2), valve materials shall be in accordance with table I. Alternate materials may be considered in lieu of those specified in table I, provided they are approved by NAVSEA.



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TABLE I. List of materials.

| Name of part | Materials | Specification or standard | Remarks |
|---|---|---|---|
| Body<br>Bonnet<br>Yoke<br>Packing gland | Valve bronze | ASTM B61 (UNS C92200)<br>For sizes under 16 inches only | 1/, 2/ |
| | Nickel aluminum bronze | MIL-B-24480,<br>ASTM B148, alloy 958<br>(UNS C95800)<br>All sizes | 1/, 2/, 3/, 4/ |
| Threaded fasteners | Silicon bronze | ASTM B98/B98M<br>(UNS C65100) | ANSI/ASME B18.2.1 and ANSI/ASME B18.2.2<br>1/ |
| | Nickel-copper alloy | QQ-N-281,<br>ASTM B164 (UNS N04400) | |
| Stem<br>Seat ring<br>Disc | Nickel aluminum bronze | MIL-B-24480,<br>ASTM B148 alloy 958<br>(UNS C95800) | 1/, 2/, 3/, 4/ |
| | Wrought nickel-copper alloy | QQ-N-281,<br>ASTM B164 (UNS N04400)<br>QQ-N-286,<br>SAE AMS4676 (UNS N05500) | |
| | Cast nickel-copper alloy | MIL-C-24723 composition M-25S,<br>SAE AMS4892 (UNS N04019) | |
| Yoke bushing | Manganese bronze | ASTM B584 (UNS C86500) | Operating nut |
| Bonnet gasket | Rubber sheet | MIL-PRF-1149, class 2 | For oil service |
| | Fluorocarbon | SAE AMS7276 | |
| | Ethylene propylene rubber | MIL-DTL-22050 | For steam or water service |
| Stem packing | Polytetrafluoroethylene (PTFE) | ASTM D1710 | Except seawater |
| | Graphitic | MIL-P-24503 | |
| Hand wheel | Aluminum alloy | 803-1385620 | |

NOTES:

1/ Table I contains the commercial and military specifications and standards and, in addition, contains the corresponding Department of Defense (DoD)-adopted SAE HS 1086 Unified Numbering System (UNS) designation. Reference to a specific UNS designation will permit the use of any one of the many alloy specifications and standards listed in SAE HS 1086 for that UNS number.

2/ Valves for seawater service shall be provided with nickel-copper alloy trim, such as discs, seats, and stem guides, as applicable.

3/ For valves where low permeability (nonmagnetic) material is required, QQ-N-286 shall be used for valve trim.

4/ Temper anneal at 1,350±50 °F for 6 hours. Required for seawater service (see ASTM B148).



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3.2.1 Prohibited materials. The following materials shall be prohibited: magnesium and its alloys, cadmium plating, asbestos, and any radioactive materials. The use of cadmium-plated parts and fasteners, including washers, is also prohibited. Lead and its compounds shall not be used where they can come in contact with seawater feed, permeate, or freshwater. The lead content in bronze alloys shall not exceed levels allowed by ASTM B61. All materials that may come in contact with potable freshwater shall be NSF/ANSI 61 approved for drinking water contact.

3.2.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Service ratings. Service ratings shall be as specified in table II.

TABLE II. Service ratings.

| Type | Class | NPS (inches) | Pressure max ($lb/in^2$) gauge | Temperature max (°F) |
|---|---|---|---|---|
| I | 2 | 2½ to 12, inclusive<br>250 $lb/in^2$ max | See 803-2177917 | |
| | 3 | 2½ to 12, inclusive<br>400 $lb/in^2$ max | 400 | 150 1/ |
| | | | 360 | 200 |
| | | | 320 | 250 |
| | | | 280 | 300 |
| | | | 240 | 350 |
| | | | 200 | 400 |
| | | | 150 | 425 |
| | 4 | 14 to 46, inclusive | 50 | 150 1/ |
| II | -- | 14 to 46, inclusive | 50 | 150 1/ |

NOTE:

1/ The service pressure rating at temperatures below 150 °F shall be the same as the rating shown for 150 °F. Valves used at lower temperatures shall conform to the requirements of the applicable condition under which they are used.

3.3.1 Type 1, class 2 valves. Type 1, class 2 valves shall be constructed in accordance with 803-2177917.

3.4 Construction. Construction of the valves shall be as specified in 3.4.1 through 3.4.18.

3.4.1 Walls and bodies. The walls of valve bodies and bonnets shall be curved surfaces. Flat surfaces shall not be permitted. Ribbing shall be provided as necessary to prevent distortion. Provisions shall be made on type II valves so that the valve body and bonnet may be assembled in one relative position only.

3.4.2 Packing. Valves shall be constructed so that the valve stem may be packed when the valve is in the fully opened position (backseated) with the maximum working pressure in the valve.

3.4.3 Flanges. Valves shall have flanged ends. Flanges shall be in accordance with MIL-PRF-20042 for pressure rating and drilling. Drilled holes shall straddle and be equally spaced from the vertical centerline and from each other. Where special drilling is required, flanges may be supplied blank or drilled special as specified (see 6.2).



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3.4.4 Spot facing. Flanges shall be spot-faced or finished on the back parallel to the flange face within 1 degree for the bolt heads and nuts. The spot face shall not cut more than ½ inch below the minimum plate thickness. Eccentricity of the spot face diameter with the bolt hole shall be minimized to provide facing for the bolt head. The diameter for spot facing shall be equal to or greater than 110 percent of the bolt head corner to corner dimension. Rims need not be furnished provided they are cast smooth and true.

3.4.5 Flow area. When the valve is fully open, there shall be an unobstructed flow. The area at any point within the valve in the line of flow shall be not less than that of the bore of the valve (see 3.9.1 and 3.9.2).

3.4.6 Disc guide ribs. The valves shall be provided with disc guide ribs in the body, which shall maintain alignment of the gate during operation. The ribs shall be accurately machined unless they are cast smooth and true.

3.4.7 Bonnets. Bonnets for all valves shall be flanged. The bonnets shall be bolted with the bolt studs threaded the full length and fitted with a nut at each end or bolted with headed bolts having a nut at one end. Bonnet flange shape shall be either circular or elliptical.

3.4.8 Stuffing boxes. Stuffing boxes shall be of a depth approximating the stuffing box bore and shall have not less than four full rings of square packing.

3.4.9 Threads. Each stem shall be threaded with National ACME form threads for valve operation in accordance with FED-STD-H28 or ANSI/ASME B1.5. Hand wheel hold-down threads shall be in accordance with FED-STD-H28. The valve shall close when facing the hand wheel and the hand wheel is turned in the clockwise direction.

3.4.10 Valve hand wheels. Hand wheel sizes shall be in accordance with 3.9.1 and 3.9.2. The hand wheel, stem, and associated parts shall be designed to withstand the maximum forces specified in table III.

TABLE III. Maximum allowable forces and torques rounded to whole numbers.

| Hand wheel diameter (inches) | Maximum allowable tangential force to seat valve (pounds) 1/ | Maximum allowable torque to seat valve (foot-pounds) 1/ | Maximum allowable tangential force to open and operate valve (pounds) | Maximum allowable torque to open and operate valve (foot-pounds) |
|---|---|---|---|---|
| 2 | 90 | 8 | 40 | 3 |
| 3 | 98 | 12 | 40 | 5 |
| 4 | 106 | 18 | 40 | 7 |
| 5 | 112 | 23 | 40 | 8 |
| 6 | 118 | 30 | 40 | 10 |
| 7 | 121 | 35 | 40 | 12 |
| 8 | 124 | 41 | 40 | 13 |
| 9 | 127 | 48 | 40 | 15 |
| 10 | 130 | 54 | 40 | 17 |
| 11 | 133 | 61 | 40 | 18 |
| 12 | 135 | 68 | 40 | 20 |
| 14 | 138 | 81 | 40 | 23 |
| 16 | 141 | 94 | 40 | 27 |
| 18 | 144 | 110 | 40 | 30 |
| 21 | 147 | 131 | 40 | 35 |
| 24 | 150 | 150 | 40 | 40 |



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TABLE III. Maximum allowable forces and torques rounded to whole numbers – Continued.

| Hand wheel diameter (inches) | Maximum allowable tangential force to seat valve (pounds) 1/ | Maximum allowable torque to seat valve (foot-pounds) 1 | Maximum allowable tangential force to open and operate valve (pounds) | Maximum allowable torque to open and operate valve (foot-pounds) |
|---|---|---|---|---|
| 27 | 150 | 169 | 40 | 45 |
| 30 | 150 | 188 | 40 | 50 |
| 36 | 150 | 225 | 40 | 60 |

NOTE:
1/ Values specified to seat a valve only apply to the maximum force or torque that shall be applied in conjunction with the test for seat leakage (see 4.5.5.3). These values do not represent the maximum allowed force or torque required to operate the valve.

3.4.11 Thread lubricant. National ACME form threads shall be lubricated with grease in accordance with DOD-G-24508.

3.4.12 Operating torque. Maximum torque required to open the valve (breakaway torque) and operate the valve (running torque) shall not exceed the values in table III (see 4.5.8).

3.4.13 Valve discs. Valve discs shall be wedge type, double-faced, one piece, either of solid cast or forged nickel-copper alloy or nickel-aluminum (Ni-Al) bronze with an insert of nickel-copper alloy (cast, forged, or sheet) securely silver brazed to the disc to form the seating surface. Fabrication and inspection of the silver brazing shall be in accordance with 0900-LP-001-7000. The finished thickness of the disc shall be as specified in table IV.

TABLE IV. Finished thickness of disc.

| Valve nps (inches) | Minimum disc thickness (inch) |
|---|---|
| 2½ to 6 | 1/8 |
| 7 to 12 | 5/32 |
| 14 to 22 | 3/16 |
| 24 to 30 | 1/4 |
| 32 to 46 | 3/8 |

3.4.14 Valve actuators. When specified (see 6.2), either manual gear (see 3.4.14.1) or electric (see 3.4.14.2) actuators shall be provided. Valves shall be constructed to assure means of mounting actuators on the valves.

3.4.14.1 Manual gear actuators. Manual gear actuators shall be provided in accordance with MSS SP-91.

3.4.14.2 Electric valve actuators. Type I, class 4 and type II valves shall be compatible with electric valve actuators in accordance with DOD-V-24657.

3.4.15 Discs. Discs for type II valves shall be self-draining.

3.4.16 Welding and allied processes. Welding and allied processes shall be in accordance with S9074-AR-GIB-010/278.

3.4.17 Valve indicator. Type II valves shall be provided with an indicator to show gate position, open, closed, and intermediate positions.



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3.4.18 Valve operation. Valves shall operate in excess of 2,000 cycles (see 4.5.4).

3.5 Shock resistance. The complete valve assembly, principal unit, shall withstand high-impact shock in accordance with grade A, class I, type A of MIL-DTL-901.

3.6 Vibration. Valves provided in accordance with this specification shall pass the type I vibration requirements in accordance with MIL-STD-167-1.

3.7 Seat leakage. Valves provided in accordance with this specification shall pass a test for seat leakage (see 4.5.5.3).

3.8 Body marking and identification plates. Body marking and identification plates shall be as specified in 3.8.1 and 3.8.2.

3.8.1 Body marking. The following markings shall be integrally cast on the valve body:

a. Manufacturer's name or trademark.

b. Valve nps in inches.

c. Valve pressure rating in $lb/in^2$.

d. MIL-DTL-1189.

3.8.2 Identification plate. Valves shall be provided with a permanently attached identification plate in accordance with MIL-DTL-15024 and MIL-P-15024/5, type A, B, C, or D. The identification plate shall contain the following:

a. Manufacturer's name or trademark.

b. PIN (see 1.3).

c. Size of valve in inches and rating in $lb/in^2$, type and class.

d. Body and bonnet material composition.

e. Valve trim material (stem, disc, seat).

f. Manufacturer's identification number (Commercial and Government Entity [CAGE] code).

3.9 Valve dimensions and weights. Valve dimensions and weights shall be as specified in 3.9.1 and 3.9.2.

3.9.1 Type I, class 3 valves. Dimensions and weights for type I, class 3 valves shall be as specified in table V for direct hand wheel to stem actuated valves.

TABLE V. Dimensions and weights of type I, class 3 valves.

| Valve size and bore diameter (inches) | Face to face (inches) 1/ | Maximum height to top (inches) | Maximum hand wheel diameter (inches) | Maximum weight (pounds) |
|---|---|---|---|---|
| 2½ | 8 | 15 | 9 | 47 |
| 3 | 8½ | 16 | 10 | 60 |
| 4 | 9½ | 18½ | 11 | 85 |
| 5 | 10¼ | 21½ | 12 | 115 |
| 6 | 10¾ | 24½ | 16 | 160 |
| 8 | 11¾ | 30 | 21 | 280 |



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TABLE V. Dimensions and weights of type I, class 3 valves – Continued.

| Valve size and bore diameter (inches) | Face to face (inches) 1/ | Maximum height to top (inches) | Maximum hand wheel diameter (inches) | Maximum weight (pounds) |
|---|---|---|---|---|
| 10 | 13½ | 36½ | 21 | 550 |
| 12 | 14½ | 39½ | 22 | 662 |

NOTE:

1/ The tolerance is ±1/64 for face-to-face dimensions less than 6 inches, and ±1/32 for face-to-face dimensions 6 inches and greater.

3.9.2 Type I, class 4 and type II valves. Type I, class 4 and type II valve dimensions and weight shall be in accordance with table VI for direct hand wheel to stem actuated valves.

TABLE VI. Dimensions and weights for type I, class 4 and type II valves.

| Valve size and bore diameter (inches) | Face to face (inches) 1/ | Maximum height to top 2/ (inches) | Maximum hand wheel diameter (inches) | Maximum weight (pounds) |
|---|---|---|---|---|
| 14 | 11½ | 42 | 18 | 1,650 |
| 16 | 11½ | 46 | 18 | 1,700 |
| 18 | 12 | 50 | 18 | 1,750 |
| 20 | 12 | 54 | 18 | 1,850 |
| 22 | 12½ | 56 | 21 | 1,950 |
| 24 | 12½ | 62 | 21 | 2,150 |
| 26 | 13 | 65 | 24 | 2,350 |
| 28 | 13½ | 70 | 24 | 2,600 |
| 30 | 14 | 74 | 24 | 2,900 |
| 32 | 15 | 78 | 24 | 3,200 |
| 34 | 15 | 84 | 24 | 3,600 |
| 36 | 15 | 87 | 24 | 4,000 |
| 38 | 16 | 91 | 24 | 4,400 |
| 40 | 17 | 94 | 24 | 4,800 |
| 42 3/ | 16 | 175 | 30 | 5,000 |
| 46 | 17 | 195 | 33 | 6,500 |

NOTES:

1/ The tolerance is ±1/32 for face-to-face dimensions 24 inches and less and ±1/16 for face-to-face dimensions greater than 24 inches.

2/ Centerline of the run of the valve to the top of the stem protector.

3/ Sizes 42 inches and larger shall be as approved by NAVSEA.

3.10 Special tools. Valves shall be constructed so that special tools shall not be required for installation and maintenance. Special tools are defined as those tools not listed in the GSA Global Supply Catalog.



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3.11 Workmanship. Casting surfaces shall be free from visual defects specified in S9074-AR-GIB-010/278. Unacceptable visual surface discontinuities shall be removed, and their removal shall be verified by visual examination of the resultant cavities. Castings shall not be repaired, plugged, impregnated, brazed, burned in, or peened except as approved.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on bronze gate valves when a first article sample is required (see 3.1). This inspection shall include the examination of 4.4 and the tests of 4.5.1 through 4.5.8. Valve test sizes shall be as specified in table VII. One valve of each type and class shall be subjected to first article inspection.

TABLE VII. Valve test sizes.

| Size tested (inches) | Approves these sizes (inches) |
|---|---|
| 4 | 2 ½ to 4, inclusive |
| 8 | 5 to 8, inclusive |
| 10 | 9 to 10 |
| Over 10 | Each size shall be tested |

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5.5 and 4.5.8. Conformance inspection shall be performed on every lot of valves acquired under this specification.

4.3.1 Sampling. Sampling shall be as specified in 4.3.1.2.

4.3.1.1 Lot. A lot shall consist of all valves of the same type, class, and size offered for delivery at one time for purpose of examination and tests.

4.3.1.2 Sampling for visual and dimensional examination. A random sample of valves shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and the acceptance quality limit (AQL) specified (see 6.2).



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TABLE VIII. Examination and inspections.

| Characteristic | First article | Conformance | Requirement | Verification |
|---|---|---|---|---|
| Visual and dimensional examination | X | X | 3.2, 3.4, 3.8, 3.9, 3.10, 3.11 | 4.4 |
| Shock | X | -- | 3.5 | 4.5.1 |
| Vibration | X | -- | 3.6 | 4.5.2 |
| Valve actuators | X | -- | 3.4.14 | 4.5.3 |
| Operation | X | -- | 3.4.18 | 4.5.4 |
| Hydrostatic strength and seat leakage | X | X | 3.7 | 4.5.5 |
| Silver brazing | X | -- | 3.4.13 | 4.5.6 |
| Welding | X | -- | 3.4.16 | 4.5.7 |
| Operating torque | X | X | 3.4.12 | 4.5.8 |

4.4 Examinations. Examinations shall be as specified in 4.4.1 and 4.4.2.

4.4.1 Examinations of materials. Materials used in the manufacture of valves shall be examined to determine conformance to 3.2 and table I, as applicable.

4.4.2 Visual and dimensional examination. Each sample valve shall be examined to determine conformance to all requirements of this specification not involving tests. Each sample valve that fails to meet these requirements shall be rejected.

4.5 Tests. Tests shall be as specified in 4.5.1 through 4.5.7.

4.5.1 Shock. The valve shall be subjected to the grade A shock test in accordance with MIL-DTL-901 to determine conformance to 3.5. Valves shall be hydrostatically pressurized internally to the seat leakage pressure specified in table IX with the valve in the full open position pressurized throughout and in the full closed position pressurized on the inlet side. Testing medium shall be clean tap water with no additive other than cutting oil at temperature not less than 40 °F or exceeding 150 °F. Momentary malfunction at the time of impact blow shall be permitted. Permanent deformation, misalignment, or functional impairments shall be cause for rejection. Permissible seat leakage shall be 10 cubic centimeters ($cm^3$) per hour of water per inch of nps.

TABLE IX. Test pressures in $lb/in^2$.

| Type | Class | Hydrostatic | |
|---|---|---|---|
| | | Strength and porosity | Seat leakage |
| I | 2 | See 803-2177917 | |
| | 3 | 600 | 400 |
| | 4 | 75 | 50 |
| II | -- | 75 | 50 |

4.5.2 Vibration. The valve shall be subjected to the type I vibration test in accordance with MIL-STD-167-1 to determine conformance to 3.6.

4.5.3 Valve actuators. When valves are provided with electric actuators, testing shall be in accordance with DOD-V-24657.



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4.5.4 Operational test. Valves shall be cycled a minimum of 2,000 times pressurized to the valve design pressure. During the test, there will be a differential pressure equal to the valve design pressure existing across the valve at the beginning of each opening cycle. Seat leakage test in accordance with 4.5.5.3 shall be performed at the start of the test, again after 1,000 cycles, and after 2,000 cycles. Test media shall be as specified in 4.5.1. There shall be a flow of water through the valve when it is open. The quantity of flow will depend upon the test facility. Valves shall open against a differential pressure equal to at least 90 percent of the rated pressure specified in 3.3. The time of closure for each cycle shall not exceed 35 seconds.

4.5.5 Hydrostatic pressure and seat leakage. Hydrostatic pressure and seat leakage tests shall be as specified in 4.5.5.1 through 4.5.5.4.

4.5.5.1 Test conditions. Each valve in a lot shall be hydrostatically tested for structural integrity and seat leakage using commercially available tap water at temperatures not less than 40 °F and not more than 150 °F and shall contain no additives other than cutting oil. Test fixtures shall be adjusted to permit valve body expansion and elongation caused by internal pressure.

4.5.5.2 Hydrostatic strength test. Each valve shall be hydrostatically pressurized to the pressure specified in table IX. The valve shall be in the open position, but not backseated. External forces or moments shall not be applied during tests. Pressure shall be maintained until a thorough visual inspection has been conducted but not less than 1 minute. Any weeping or leakage during pressurizing or permanent deformation after pressurizing shall be cause for rejection.

4.5.5.3 Hydrostatic seat leakage test. Each valve shall be tested with the disc in the shut position by pressurizing each side with the side opposite the pressure open for inspection in each case. The valve shall be seated using the force as specified in table III as the maximum allowable hand wheel rim force. Test pressure as specified in table IX shall be maintained until a visual inspection has been conducted. If no leakage occurs after 1 minute, test may be discontinued. If leakage does occur, the test shall be continued for sufficient time to accurately determine the rate of leakage. Leakage shall not exceed 10 $cm^3$ per hour of water per inch of nps.

4.5.5.4 Steam test. Two leak tests shall be conducted using steam at 150 and 50 $lb/in^2$ gauge saturated steam. The tests shall be conducted for main seats following the procedure outlined in 4.5.5.3. The leakage rate shall not exceed 10 $cm^3$ of condensate per inch nps. Valves shall be cycled 25 times with seat leakage measurements taken before and after cycling.

4.5.6 Silver brazing. Silver brazing shall be inspected in accordance with 0900-LP-001-7000 to determine conformance to 3.4.13.

4.5.7 Welding. Welds shall be inspected in accordance with S9074-AR-GIB-010/278 to determine conformance to 3.4.16.

4.5.8 Operating torque. Each valve shall be cycled once (closed-open-closed) and the torque required to open the valve (breakaway torque) and operate the valve (running torque) shall be measured. The torque values shall not exceed those specified in 3.4.12.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.



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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Type I, class 2 and 3 valves are intended to be used primarily for water, oil, gas, and steam service onboard naval ships. Type I, class 4 and type II valves are intended for use on seawater services as main injection and overboard discharge valves onboard naval ships. Current stock of bronze gate valves may be used until depleted; future requisitions for bronze gate valves should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, and size (see 1.2).
c. Quantity.
d. When first article is required (see 3.1).
e. Material composition (see 3.2 and table I).
f. Flange drilling (see 3.4.3).
g. When valve actuator is required (see 3.4.14).
h. AQL of 1.0 (see 4.3.1.2).
i. Packaging requirements (see 5.1).
j. Recommended documentation (see 6.3).

6.3 Recommended documentation. The following list of documentation is recommended to be requested as part of the acquisition of any valve(s) under this specification (see 6.2):

a. Engineering drawings
b. Materials listing
c. Installation drawings
d. Technical manuals
e. Test reports
f. On-board repair parts list

6.4 Subject term (key word) listing.

Bonnets
Electric actuators
Flanges
Hand wheels
Seat rings
Stuffing boxes

6.5 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.



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CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4820-2019-009)

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